Pre-A Q3LOCK Gibbs-L2, Connected QPS, and Common-Alpha Boundary Checkpoint

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1 Purpose and scope

This single gate-level checkpoint source synthesizes EXP-000815 and advances reusable result R-167 v2.3 without changing a claim card or tier. The result is additive, T0, and claim_bearing: false. Its stable result identifier is

PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT.

R-167 v2.3 preserves every v2.2 and earlier theorem, route boundary, negative result, and issued source/PDF pair. It closes three additional scoped children. First, the registered periodic two-orientation history corridor gives mesh-uniform hard-coordinate-cutoff removal for the declared split implementers in a two-sided Gibbs Hilbert–Schmidt seminorm. Second, a deterministic connected-set count converts a separately assumed geometric coefficient envelope into the required QPS interaction norm and a dominated cutoff criterion. Third, the actual second-order onsite-resolvent coefficient is connected, obeys an explicit QPS bound, and converges under a declared resolvent-compatible parity-preserving Ritz/form cutoff contract.

An exact one-sided UHF fixture separately rejects the inference that forward local stabilization of finite-volume automorphisms automatically yields a surjective limiting automorphism or point-norm Cauchy inverses. It is a route boundary, not a nonexistence theorem for Q3LOCK dynamics.

This is an R-167 v2.3-only synthesis. EXP-000816 is recorded here only as a parked route-authority audit: it creates no M2 candidate, theorem, version, prediction, or freeze. R-168 v1.3 remains historical and is not reissued. No current R-168 verification total is part of this result.

The complete combined R-167 v2.2 / R-168 v1.3 v1.1 source/PDF pair remains historical evidence. No per-lemma or intermediate PDF belongs to the present stage. The v1.2 sibling PDF is now issued exactly once, after the proof, formal-authority, primary, non-importing independent, integrated, generated-surface, source-form, freshness, dual-extraction, strict-release, and visual-review gates passed.

2 Closed children, negative authority, and surviving parents

The three children closed by EXP-000815 are exactly

1. PA-CP1-ST8-Q3LOCK-REGISTERED-PERIODIC-SPLIT-IMPLEMENTER-TWO-SIDED-GIBBS-L2-HARD-CUTOFF-REMOVAL;
2. PA-CP1-ST8-Q3LOCK-CONDITIONAL-CONNECTED-CLUSTER-GEOMETRIC-QPS-NORM-ENVELOPE;

3. PA-CP1-ST8-Q3LOCK-SECOND-ORDER-CONNECTED-ONSITE-RESOLVENT-QPS-NORM-AND-RITZ-CUTOFF.

The new scoped negative authority is exactly

NG-2026-08-11-PRE-A-ST8-Q3LOCK-FORWARD-LOCAL-AUTOMORPHISM-LIMIT-AUTOMATIC-SURJECTIVITY-AND-INVERSE-CAUCHY.

The following load-bearing parents and successors remain **OPEN**:

1. PA-CP1-ST8-Q3LOCK-LOCAL-STRICT-ALL-EXHAUSTION-TWO-ORIENTATION-HISTORY-COMMON-ALPHA;
2. PA-CP1-ST8-Q3LOCK-CONNECTED-RANK-TWO-OSCILLATOR-ELIMINATION-QPS-NORM-AND-CUTOFF-COMPATIBILITY;
3. PA-CP1-ST8-Q3LOCK-INFINITE-DIMENSIONAL-RANK-TWO-BAND-BLOCK-DIAGONALIZATION-AND-TWO-PHASE-QPS;
4. PA-CP1-ST8-Q3LOCK-BROKEN-SECTOR-GNS-GAP-COERCIVITY;
5. PA-ROUND1-EVIDENCE-ROLE-AND-MINIMUM-MANIFEST-FREEZE.

The M2 successor gates reviewed by EXP-000816 also remain open:

1. PA-M2-CI8-PHYSICAL-RESPONSE-CHANNEL-AND-ERROR-BOUND;
2. PA-M2-SUCCESSOR-SUBSTANTIVE-COMPACT-ACTION-BACKGROUND-PROBE-AND-WINDING-LAW;
3. PA-M2-SUCCESSOR-ORDERED-STATE-PHYSICAL-MODE-AND-RESPONSE-LIMIT;
4. PA-M2-SUCCESSOR-SIX-TERM-CRITICAL-ESTIMAND-ERROR-BUDGET.

The three closed children are reusable local results. Their conjunction does not close any item in either open list.

3 Two-sided Gibbs-L2 cutoff removal for split implementers

Fix one registered finite periodic Q3 volume, one source in the registered compact source family, and one declared finite split mesh. Let P be a declared partial or reverse split implementer, let P_L be its hard-coordinate-cutoff counterpart, and let ρ be the full finite-volume Gibbs density. For an implementer difference X, define

$$\|X\|_{\rho,\#1} := \|X\rho^{1/2}\|_2 + \|\rho^{1/2}X\|_2. \tag{3.1}$$

The two Hilbert–Schmidt legs require opposite Duhamel orderings. Telescoping the forward and reverse histories separately, while using unitarity for every unchanged factor, gives

$$\|(P-P_L)\rho^{1/2}\|_2 \le \frac{T}{\hbar}\sqrt{C_+}, \qquad \|\rho^{1/2}(P-P_L)\|_2 \le \frac{T}{\hbar}\sqrt{C_-}. \tag{3.2}$$

Here the total absolute bond time is at most T. The bound is uniform over the registered periodic volume, compact source, declared split mesh, tested translate, and every declared partial or reverse history. Adding (3.2) and using the two-variable Cauchy inequality yields

$$\|P - P_L\|_{\rho,\#1} \le \frac{T}{\hbar}(\sqrt{C_+} + \sqrt{C_-}) \le \frac{T}{\hbar}\sqrt{2(C_+ + C_-)}. \tag{3.3}$$

The v2.2 paired-orientation authority is

$$C_+ + C_- \le 2916 c^2 M_{20} R^6 L^{-16}. \tag{3.4}$$

Since $2916 = 54^2$, the choice $L = R^{2/5}$ gives

$$\boxed{\|P - P_L\|_{\rho,\#1} \le \frac{54\sqrt{2}\,|c| T \sqrt{M_{20}}}{\hbar} R^{-1/5}.} \tag{3.5}$$

The squared rate constant before taking the Hilbert–Schmidt square root is 5832. At $R = 32$, $L = 4$, the dimensionless rate coefficient is $27\sqrt{2}$; at $R = 3125$, $L = 25$, it is $54\sqrt{2}/5$.

Equation (3.5) is an implementer estimate only. It inserts no arbitrary bounded observable between an implementer difference and either Gibbs-weighted leg. It proves neither arbitrary-observable conjugation convergence nor an all-shape thermodynamic automorphism.

4 Conditional geometric connected envelope to QPS norm

Let the host graph have maximum degree z, fix a neighbor ordering, and let $\Phi(X)$ vanish unless X is finite and connected. Assume the separate geometric input

$$\|\Phi(X)\| \le A\kappa^{|X|}, \qquad r := z^2 e^a \kappa < 1. \tag{4.1}$$

For every connected n-set containing a fixed root x, choose a canonical rooted spanning tree and traverse it by deterministic depth-first search. The walk has length $2(n-1)$ and recovers the visited vertex set. It therefore injects the connected sets into at most $z^{2(n-1)}$ walks. Since $\operatorname{diam}(X) \le n - 1$,

$$\begin{aligned} \sup_x \sum_{X \ni x} |X| e^{a\,\operatorname{diam}(X)} \|\Phi(X)\| &\le A\kappa \sum_{n \ge 1} n (z^2 e^a \kappa)^{n-1} \\ &= \frac{A\kappa}{(1-r)^2}. \end{aligned} \tag{4.2}$$

Cutoff convergence requires an additional uniform difference envelope. If

$$\|\Phi_M(X) - \Phi(X)\| \le A_M \kappa^{|X|}, \qquad A_M \longrightarrow 0, \tag{4.3}$$

uniformly over the declared boundaries, compact sources, and translates, then

$$\|\Phi_M - \Phi\|_a \le \frac{A_M \kappa}{(1-r)^2} \longrightarrow 0. \tag{4.4}$$

For the exact fixture $z = 6$, $e^a = 2$, $\kappa = 1/144$, and $A = 1/100$, one has $r = 1/2$ and the bound in (4.2) is $1/3600$. With $A_M = A/M$, one has $A_{10} = 1/1000$ and the cutoff bound is $1/36000$.

The counting theorem uses only maximum degree and a deterministic ordering. The conclusion remains conditional because no all-order Q3 oscillator calculation in this checkpoint derives either (4.1) or (4.3).

5 Actual second-order onsite-resolvent connected coefficient

Let P be the product low-band projection, $Q = 1 - P$, and let the exact onsite high-sector operator satisfy

$$K = \sum_x k_x \geq \Gamma Q, \qquad T_e = QB_eP, \qquad \|T_e\| \leq \epsilon. \tag{5.1}$$

The actual second-order onsite-resolvent coefficient is

$$\Psi^{(2)} = -\sum_{e,f} T_e^* (QKQ)^{-1} T_f. \tag{5.2}$$

The onsite sum K preserves exact high-support sectors. Starting from the all-low space, T_f creates high support only inside f. If e and f are disjoint, T_e^* cannot remove that support and the final low projection kills the vector. Thus every disjoint ordered pair vanishes. Every remaining ordered coefficient has norm at most ϵ^2/Γ.

For a fixed root site, there are at most z diagonal pairs $e = f$ whose union contains the root. There are at most $3z(z-1)$ ordered off-diagonal overlapping pairs whose union contains it. Diagonal supports have size two and diameter at most one; off-diagonal supports have size at most three and diameter at most two. Therefore

$$\boxed{\|\Psi^{(2)}\|_a \leq C_a \frac{\epsilon^2}{\Gamma}, \qquad C_a = 2ze^a + 9z(z-1)e^{2a}.} \tag{5.3}$$

At $z = 6$ and $e^a = 2$, the diagonal and ordered off-diagonal counts are 6 and 90, respectively, and

$$C_a = 12e^a + 270e^{2a} = 1104. \tag{5.4}$$

With $\epsilon = 1/100$ and $\Gamma = 2$, (5.3) is 69/1250. In the registered v2.1 corridor, $\epsilon = O(N^{-3})$ and Γ is of order N^2, so the second-order QPS norm is $O(N^{-8}) = o(J)$. This exponent follows from those registered parameter scalings; it is not inferred from spatial dimension.

For cutoff removal, let Π_M be parity-preserving local Ritz/form projections that contain P and commute with the onsite resolvent. Define

$$D_e = (QKQ)^{-1/2} T_e, \qquad D_{e,M} = (QKQ)^{-1/2} \Pi_M T_e, \qquad \tau_M = \sup_e \|(1 - \Pi_M) T_e\|. \tag{5.5}$$

Assume $\tau_M \to 0$ uniformly over the registered compact source family. Then

$$\|D_e\|, \|D_{e,M}\| \leq \frac{\epsilon}{\sqrt{\Gamma}}, \qquad \|D_e - D_{e,M}\| \leq \frac{\tau_M}{\sqrt{\Gamma}}. \tag{5.6}$$

Subtracting the two Gram products and repeating the connected-pair count gives

$$\|\Psi_M^{(2)} - \Psi^{(2)}\|_a \leq \frac{2C_a \epsilon \tau_M}{\Gamma} \longrightarrow 0. \tag{5.7}$$

At $\tau_{10} = 1/1000$, the exact fixture bound is 69/6250. The uniform Gram-tail hypothesis and resolvent commutation are load-bearing additions. Ritz eigenvalue monotonicity alone does not prove (5.7). No higher connected coefficient, all-order convergence radius, quasi-local intertwiner, phase transfer, or broken-sector GNS gap follows.

6 Forward local stabilization does not control inverses

Let

$$\mathcal{A} = \bigotimes_{j\geq 1} M_2 \tag{6.1}$$

be the one-sided UHF algebra. For each N, let α_N cyclically send sites $1,\dots,N$ to $2,\dots,N,1$ and fix all later sites. Each α_N is an automorphism. If A is supported in the first m sites, then for every $N > m$,

$$\alpha_N(A) = 1 \otimes A. \tag{6.2}$$

Thus the forward automorphisms stabilize exactly on every local observable and converge pointwise on the quasi-local algebra to the unilateral right shift $\sigma(A) = 1 \otimes A$. The range of σ has identity in its first tensor factor. Hence the first-site Pauli Z_1 is not in the range and σ is a proper injective unital endomorphism, not an automorphism.

In the inverse direction,

$$\alpha_N^{-1}(Z_1) = Z_N, \qquad \|Z_N - Z_M\| = 2 \quad (N \neq M). \tag{6.3}$$

A product eigenvector with opposite Z signs at sites N and M attains the upper bound two. The exact verifier fixture checks forward support $(1,2,3) \mapsto (2,3,4)$ and the inverse pair $(N,M) = (3,5)$.

This proves only the named implication failure. It does not reject a Q3LOCK common alpha established with independent forward and inverse Cauchy control, surjectivity, group law, generator identification, and phase-KMS quotients.

7 Parked M2 successor audit and historical firewall

The contemporaneous audit `EXP-000816` asks whether current M2-v0 and R-168 authorities license automatic promotion to a complex or gauge-covariant compact-action successor. They do not. A complex or compact field, nontrivial charge and action, background gauge or bundle data, a covariant derivative, quadratic contact, winding quotient, ordered state, and response law are substantive new model choices rather than consequences of R-168 v1.3.

The verdict is `parked`, not advanced and not a no-go. Any future successor requires explicit operator authorization as a separately versioned T0 candidate, a complete D00–D09 rerun, and a genuinely prospective external holdout distinct from the already-visible helium target. This audit creates no candidate, map, prediction, target, freeze, score, selection, or new R-168 theorem. R-168 remains v1.3. Its issued v1.1 combined checkpoint remains historical and is not reissued by this R-167-only source/PDF lifecycle.

8 Devil's-advocate review

1. **UPHELD as a boundary – arbitrary observable insertion:** the Gibbs-L2 implementer estimate contains no observable between either weighted leg and $P - P_L$. A bounded half-modular context or a new argument is still required for conjugations.

2. **DISMISSED – hidden cubic geometry in the connected count:** the rooted depth-first encoding uses only the declared maximum degree z and a fixed neighbor order. The coefficient envelope, not the count, is the conditional input.

3. **UPHELD – pointwise cutoff convergence:** pointwise convergence does not control a supremum-sum QPS norm. The uniform difference envelope with $A_M \to 0$ is stated separately.

4. **UPHELD – Ritz monotonicity:** decreasing Ritz eigenvalues do not imply convergence of the resolvent Gram factors. Resolvent-compatible parity projections and the compact-family-uniform $\tau_M \to 0$ bound are required.

5. **UPHELD as an overclaim – second order to all order:** the exact second-order term supplies no higher connected coefficient, convergence radius, remainder estimate, quasi-local transformation, or phasewise GNS intertwiner.

6. **DISMISSED – UHF nonexistence promotion:** the fixture rejects an automatic implication from forward convergence only. It is not a Q3LOCK dynamics no-go.

7. **VALID with mitigation – numerical and provenance pins:** the three engines recompute the fixtures independently and bind their raw hashes, but their pass totals certify only the declared formulas, authorities, and scope. They do not upgrade the result tier or close a physical parent.

8. **UPHELD – automatic M2 successor:** importing a complex field, gauge structure, or visible target without explicit authorization would erase model provenance and bypass prospective validation. `EXP-000816` therefore remains parked and R-168 v1.3 remains historical.

Independent adversarial review is invited especially on the two Duhamel orientations, the factor $\sqrt{2}$ in (3.3), injectivity of the rooted-walk encoding, the $3z(z-1)$ ordered-pair count, resolvent commutation and the uniform Gram tail, the UHF inverse norm, the N^{-8} scaling provenance, and every no-overclaim boundary.

9 Reproduction and exact verification pins

The R-167 v2.3 primary, non-importing independent, and integrated scripts are

1. `codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split.py`;
2. `codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_independent.py`;
3. `codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_verify.py`.

From the repository root, run

```
Set-Location E:\Dev\TECT
$py = 'E:\Dev\TECT.venv\Scripts\python.exe'
$r167 = (
  'codes\foundations\pre_a_cp1_st8_q3lock_' +
  'local_measured_renyi_semiclassical_doublet_route_split'
)
& $py -X utf8 ($r167 + '.py')
& $py -X utf8 ($r167 + '_independent.py')
& $py -X utf8 ($r167 + '_verify.py')
```

The final outputs are primary `PASS 295/295`, non-importing independent `PASS 175/175`, and integrated `PASS 304/304`. The exact raw verification-script SHA-256 pins are

```
R-167 primary:
ed9a57a81e484d99d957de01d54ec9e078447e332fbff5b909225740efd08a40
R-167 independent:
c5cdffa7d01637024630ca98f55634793fa2b98e6dcdff6157c5ba3858ce5bb5
R-167 integrated:
049f3005de45b4a0b22d5a4188861266ede45c9c11a159bc13a080b199141341
```

The primary symbolic engine and the non-importing standard-library engine recompute the paired-history Gibbs-L2 arithmetic, the geometric-envelope sum, the second-order pair count and cutoff Gram bound, and the UHF fixture from labelled inputs. The integrated engine binds those computations to the exact exploration, result, gate, negative, strategy, generated-surface, stored-run, and checkpoint-lifecycle authorities. A failed check invalidates the total; it is never converted into a weaker pass.

Source-form validation with `build_note_pdf.py -no-compile` is not PDF issuance. At the issued stage, the exact manifest workflow is:

```
No per-lemma or intermediate PDF was issued.  One R-167 v2.3 gate-level synthesis
source/PDF pair was issued only after the primary, non-importing independent,
integrated, formal-authority, generated-surface, source-form, freshness,
dual-extraction, strict-release, and visual-review checks passed.  R-168 v1.3
remains historical and is not reissued.
```

The exact issued visual-QA record is:

```
All 8 rendered pages were reviewed at readable resolution with zero clipping,
overlap, broken equations, unreadable identifiers, black glyphs, or malformed page
transitions; pypdf and pdfplumber each extracted 8/8 nonempty pages; the build
reported OVERFULL-HBOX 0.
```

That lifecycle is now issued. The sibling PDF was built fresh relative to this source, extracted successfully by both pypdf and pdfplumber on all eight pages, and reviewed page by page from rendered PNGs. R-168 v1.3 remains historical and is not reissued.

10 Exact no-overclaim and dependency boundary

The common-alpha parent is **OPEN**. The finite-periodic, state-weighted implementer estimate does not prove arbitrary-observable convergence, $n \to \infty$ Trotter convergence, all-shape exhaustion Cauchy compatibility, surjectivity, inverse convergence, group or star/product closure, a generator, or fixed-beta phase-KMS quotients.

The connected oscillator-elimination successor is **OPEN**. The conditional envelope theorem constructs no actual all-order Q3 coefficient, and the exact second-order coefficient supplies no all-order transformation, remainder, convergence radius, or cutoff theorem.

The broader infinite-dimensional rank-two parent is **OPEN**. No boundary/source-uniform parity-equivariant quasi-local elimination or exact oscillator two-phase QPS transfer is proved.

The broken-sector spectral parent is **OPEN**. No phasewise GNS intertwiner, oscillator temporal mass, oscillator-lattice GNS gap, or physical mass gap follows from the second-order interaction norm.

The Round-1 evidence-role parent is **OPEN**. `EXP-000816` is a parked authorization audit, not a candidate or a prospective freeze. It does not alter R-168 v1.3 or authorize a complex/covariant M2 successor.

Nothing in this checkpoint proves regulator removal, continuum control, same-Hamiltonian physical-empty comparison, a below-empty sign, physical vacuum selection, C6 emergence, CP1, physical Sector A, or Pre-A closure. The registered `A5-SECTOR-A-SYNTHESIS` remains conditional at its stated scope, and the `C6-SPACETIME-SIGNATURE` claim card and tier are unchanged.

11 Result footer

```
Result ID:              PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-
                        MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT.
Reusable result:        R-167 v2.3.
Exploration:            EXP-000815.
```

```
Parked route audit:       EXP-000816; no result, candidate, or version change.
Precise statement:        Registered-periodic two-sided Gibbs-L2 hard-cutoff
                          removal for declared split implementers; conditional
                          connected geometric-envelope-to-QPS theorem; exact
                          second-order onsite-resolvent connected QPS norm and
                          resolvent-compatible Ritz/form cutoff convergence;
                          forward-limit surjectivity/inverse-Cauchy no-go.
Claim context:            C6-SPACETIME-SIGNATURE; claim_bearing false.
Closed children:          Exactly the three R-167 identifiers in Section 2.
Negative result:          Exactly the UHF implication failure in Section 2.
Open successor gates:     All-exhaustion common alpha; connected all-order
                          rank-two oscillator elimination/QPS and cutoff;
                          broader rank-two two-phase transfer; broken-sector
                          oscillator GNS gap; Round-1 evidence-role freeze.
Scope:                    T0 additive, claim-nonbearing analytic/exact result
                          in the registered fixed-beta finite-periodic compact-
                          source split-implementer scope, conditional connected
                          envelope scope, exact second-order onsite-resolvent
                          scope, and exact one-sided-UHF counterfixture scope.
Dependencies:             R-167 v2.2; EXP-000813; registered paired twentieth-
                          history corridor; onsite high-support preservation;
                          declared parity/resolvent Gram-tail cutoff contract.
Evidence grade:           ANALYTIC + EXACT + primary + non-importing
                          independent + integrated; one R-167-only gate-level
                          PDF issued after freshness, 8/8 dual extraction,
                          strict release, and page-by-page visual review passed.
Reproduction command:     Run the three commands in Section 9 from repository
                          root, primary then independent then integrated.
Expected output:          PASS 295/295; PASS 175/175; PASS 304/304.
Primary raw SHA-256:      ed9a57a81e484d99d957de01d54ec9e078447e332fbff5b909225740efd08a40.
Independent raw SHA-256: c5cdffa7d01637024630ca98f55634793fa2b98e6dcdff6157c5ba3858ce5bb5.
Integrated raw SHA-256:   049f3005de45b4a0b22d5a4188861266ede45c9c11a159bc13a080b199141341.
Falsification gate:       Any Duhamel orientation, paired-tail factor, square-
                          root/rate exponent, rooted-walk injection, connected
                          pair count, Gram/resolvent cutoff, UHF support/range/
                          inverse norm, N-scaling, assertion count, hash,
                          provenance, freshness, extraction, rendering, or
                          scope mismatch.
Tier before / after:      C6 T1 / C6 T1; no claim or tier change.
No-overclaim statement:   No arbitrary-observable or all-exhaustion common
                          alpha, all-order oscillator elimination, phasewise
                          QPS/GNS transfer, oscillator gap, new M2 successor,
                          Round-1 freeze, regulator removal, continuum,
                          physical-empty sign, C6, CP1, physical Sector A, or
                          Pre-A closure.
Next required action:     Prove forward and inverse all-shape Cauchy control,
                          surjectivity, group/generator/KMS identification;
                          independently derive and sum the all-order connected
                          oscillator interaction in one cutoff-stable QPS norm,
                          then construct the phasewise GNS intertwiner and gap.
PDF policy:               No per-lemma or intermediate PDF. The v1.2 sibling
                          PDF was issued exactly once after every proof, formal,
                          component, integrated, source, freshness, extraction,
                          visual, and release gate passed.
R-168 firewall:           R-168 v1.3 remains historical and is not reissued;
                          no current R-168 verification count is part of this
                          result footer.
```